

Lecture 10: Sept 10

Last time

- Distribution Functions

Today

- Distribution Functions

So far, we have seen finite S and finite S_X , and the definition of $\Pr_X$ is straightforward. If S_X is uncountable, we define the induced probability function, $\Pr_X$ for any set $A \subset S_X$,

$$\Pr_X(X \in A) = \Pr(\{s \in S : X(s) \in A\}).$$

This defines a legitimate probability function for which the Kolmogorov Axioms can be verified.

Distribution Functions

Distribution Functions are used to describe (and actually define) the behavior of a r.v.

Cumulative distribution function

Definition The *cumulative distribution function* or *cdf* of a random variable X , denoted by $F_X(x)$, is defined by

$$F_X(x) = \Pr_X(X \leq x), \text{ for all } x.$$

Definition The *survival function* of a random variable X , is defined by

$$S_X(x) = 1 - F_X(x) = \Pr_X(X > x).$$

Example Consider the experiment of tossing three fair coins, and let X = number of heads observed. The cdf of X is

$$F_X(x) = \begin{cases} 0 & \text{if } -\infty < x < 0 \\ \frac{1}{8} & \text{if } 0 \leq x < 1 \\ \frac{3}{8} & \text{if } 1 \leq x < 2 \\ \frac{7}{8} & \text{if } 2 \leq x < 3 \\ 1 & \text{if } 3 \leq x < \infty \end{cases}$$

Some properties of the cdf:

Let $F(x)$ be a cdf. Then

1. $0 \leq F(x) \leq 1$
2. $\lim_{x \rightarrow -\infty} F(x) = 0$

3. $\lim_{x \rightarrow \infty} F(x) = 1$
4. F is nondecreasing: if $a < b$, then $F(a) \leq F(b)$
5. F is right-continuous: $\lim_{x \downarrow b} F(x) = F(b)$, or $\lim_{x \rightarrow b^+} F(x) = F(b)$
6. $\Pr(a < X \leq B) = F(b) - F(a)$

Theorem The function $F(x)$ is a cdf if and only if the following three conditions hold:

1. $\lim_{x \rightarrow -\infty} F(x) = 0$ and $\lim_{x \rightarrow \infty} F(x) = 1$
2. F is nondecreasing: if $a < b$, then $F(a) \leq F(b)$
3. F is right-continuous: $\lim_{x \downarrow b} F(x) = F(b)$, or $\lim_{x \rightarrow b^+} F(x) = F(b)$

The cdf does not contain information about the original sample space.

Definition Two random variables X and Y are identically distributed if, for every Borel set $A \subset \mathbb{R}$, $\Pr(X \in A) = \Pr(Y \in A)$.

Example Toss a fair coin n times. The number of heads and the number of tails have the same distribution.

Theorem The following two statements are equivalent:

1. The random variables X and Y are *identically distributed*.
2. $F_X(x) = F_Y(x)$ for every x .

Types of Random Variables

Definition A random variable X can be

- *discrete*:
 - X takes on a finite or countably infinite number of values
 - $F_X(x)$ is step-wise constant
- *continuous*:
 - the range of X consists of subsets of the real line
 - $F_X(x)$ is continuous.
- *mixed*: $F_X(x)$ is piecewise continuous.

Example A random variable has cdf

$$F(x) = \begin{cases} 0 & x < 0 \\ x/2 & 0 \leq x < 1 \\ 2/3 & 1 \leq x < 2 \\ 11/12 & 2 \leq x < 3 \\ 1 & 3 \leq x \end{cases}$$

Is this a valid cdf? Is it a discrete random variable or continuous random variable or mixed?
solution:

Discrete Random Variables

Suppose a random variable X takes only a finite or countable number of values. Let the sample space of X be $S = \{x_1, x_2, \dots\}$. Then the cdf can be expressed as:

$$F(x) = \sum_{x_i \leq x} \Pr(X = x_i).$$

Definition The *probability mass function* (pmf) of a discrete random variable X is given by

$$f_X(x) = \Pr(X = x) \text{ for all } x.$$

If the sample space of X is $X = \{x_1, x_2, \dots\}$, then

$$f(x_i) = \Pr(X = x_i) = \Pr(x_{i-1} < X \leq x_i) = F(x_i) - F(x_{i-1}).$$

Example (Geometric probabilities) Suppose we do an experiment that consists of tossing a coin until a head appears. Let p = probability of a head on any given toss, and define a random variable X = number of tosses required to get a head. Then for any $x = 1, 2, \dots$,

$$\Pr(X = x) = (1 - p)^{x-1}p,$$

since we must get $x - 1$ tails followed by a head for the event to occur and all trials are independent. What is the pmf of the above Geometric distribution? What is the cdf?

solution:

Definition The *domain* of a random variable X is the set of all values of x for which $f(x) > 0$. This is also called *range*, *sample space* or *support*.

Properties of the pmf:

1. $f(x) > 0$ for at most a countable number of values x . For all other values x , $f(x) = 0$.

2. Let $\{x_1, x_2, \dots\}$ denote the domain of X . Then

$$\sum_{i=1}^{\infty} f(x_i) = 1.$$

An obvious consequence is that $f(x) \leq 1$ over the domain.